AIM

To test and compare how **different prompting patterns** (naïve/broad prompts versus basic/structured prompts) affect the **quality, accuracy, and depth** of responses generated by ChatGPT across multiple scenarios.

AI TOOLS REQUIRED

- **ChatGPT (OpenAI)**
- Internet-enabled system

EXPLANATION

Prompting patterns play a crucial role in determining the effectiveness of AI-generated responses.
This experiment evaluates how **prompt clarity and structure** influence the performance of ChatGPT by comparing:

- **Naïve Prompts**: Broad, vague, or unstructured prompts with minimal guidance.
- **Basic (Refined) Prompts**: Clear, specific, and structured prompts with explicit instructions and context.

Multiple test scenarios are used to observe how ChatGPT responds under different prompting conditions.

DEFINE THE TWO PROMPT TYPES

1. Naïve Prompt

- Broad or unstructured
- Lacks specific instructions
- Relies on AI assumptions

Example:

“Tell me about AI.”

2. Basic (Refined) Prompt

- Clear and well-structured

- Specifies task, format, and context
- Reduces ambiguity

Example:

"Explain Artificial Intelligence in simple terms, list its main types, and give one real-world example for each."

PROCEDURE

1. Identify different **test scenarios**.
2. Create both **naïve and basic prompts** for each scenario.
3. Input each prompt into **ChatGPT**.
4. Record the generated responses.
5. Evaluate outputs based on:
    - Quality
    - Accuracy
    - Depth
6. Compare results and analyze the impact of prompt clarity.

TEST SCENARIOS AND PROMPT DESIGN

Scenario 1: Creative Story Generation

Naïve Prompt:

"Write a story."

Basic Prompt:

"Write a short science fiction story (150 words) about an AI system that saves a manufacturing plant from a major breakdown. Include a clear beginning, conflict, and resolution."

Scenario 2: Factual Question

Naïve Prompt:

"What is machine learning?"

Basic Prompt:

“Define machine learning, explain how it works, and list two real-world applications in manufacturing.”

Scenario 3: Concept Summarization

Naïve Prompt:

“Summarize IoT.”

Basic Prompt:

“Summarize the concept of the Internet of Things (IoT) in 100 words, focusing on its role in smart manufacturing.”

Scenario 4: Advice / Recommendation

Naïve Prompt:

“How can factories reduce downtime?”

Basic Prompt:

“Suggest five AI-based strategies that manufacturing industries can use to reduce machine downtime, with brief explanations.”

OBSERVED OUTPUT COMPARISON TABLE

Scenario	Prompt Type	Quality	Accuracy	Depth
Creative Story	Naïve	Medium	Medium	Low
	Basic	High	High	High
Factual Question	Naïve	Medium	Medium	Low
	Basic	High	High	Medium–High
Summarization	Naïve	Low–Medium	Medium	Low
	Basic	High	High	Medium
Advice	Naïve	Medium	Medium	Low
	Basic	High	High	High

EVALUATION AND ANALYSIS

Impact of Naïve Prompts

- Responses are often **generic and shallow**
- Lack of structure leads to **missing details**
- Useful for brainstorming but not for technical tasks

Impact of Basic (Refined) Prompts

- Outputs are **clear, structured, and relevant**
- Higher **technical accuracy**
- Greater **depth and usefulness**
- Consistent performance across all scenarios

Key Observations

- ChatGPT performs **significantly better** with basic prompts
- Naïve prompts work reasonably well only for **simple or creative tasks**
- Complex tasks require **explicit instructions and context**

SUMMARY OF FINDINGS

- Prompt clarity directly affects response quality.
- Structured prompts improve **accuracy, completeness, and depth**.
- Basic prompts are essential for **academic, technical, and professional use cases**.
- Naïve prompts may be suitable for **quick ideas or casual queries**, but not for detailed analysis.

CONCLUSION

This experiment confirms that **basic (refined) prompting patterns** consistently produce superior results compared to naïve prompts. ChatGPT responds more effectively when tasks, constraints, and expectations are clearly defined. Proper prompt engineering is therefore essential for obtaining reliable and high-quality AI-generated outputs.